

A Note on Sums of Two Squares

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Abstract

We give a short, self-contained account of which primes are sums of two squares. The argument uses only the pigeonhole principle and elementary congruences.

1 Introduction

Our goal is theorem 1.1. The key input is theorem 2.2, and the statement for general integers is theorem 3.1.

Theorem 1.1. *An odd prime p is a sum of two squares if and only if $p \equiv 1 \pmod{4}$.*

2 Preliminaries

Definition 2.1. An integer n is *represented* if $n = x^2 + y^2$ for some integers x and y .

Squares are congruent to 0 or 1 modulo 4, so

$$x^2 + y^2 \not\equiv 3 \pmod{4} \tag{2.1}$$

for all integers x, y .

Lemma 2.2 (Thue). *Let p be prime and a an integer not divisible by p . There are integers x, y with $0 < |x|, |y| < \sqrt{p}$ and $ax \equiv y \pmod{p}$.*

Proof. Let $m = \lfloor \sqrt{p} \rfloor$. The $(m+1)^2 > p$ numbers $au - v$ with $0 \leq u, v \leq m$ cannot be distinct modulo p , and subtracting two equal ones gives the required x and y . $\square$

3 Proof of the main theorem

Proof of theorem 1.1. Necessity is (2.1). Conversely, if $p \equiv 1 \pmod{4}$ then -1 is a square modulo p , say $a^2 \equiv -1$. Applying theorem 2.2 to a gives

$$x^2 + y^2 \equiv x^2 + a^2 x^2 \equiv 0 \pmod{p}, \tag{3.1}$$

with $0 < x^2 + y^2 < 2p$. Hence $x^2 + y^2 = p$. $\square$

Corollary 3.1. *A positive integer is represented in the sense of theorem 2.1 if and only if every prime $q \equiv 3 \pmod{4}$ divides it to an even power.*

Proof. Combine theorem 1.1 with the identity $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$ and with (3.1). $\square$